

AI Lecture Companion Agent

Problem Solving and Problem Formulation

1. Problem Formulation

Element	Description
Initial State	The agent receives the teacher's recorded lecture audio, board pictures, slides, and teacher notes. The lecture material is not yet organized, and the agent starts processing it.
Actions	Convert lecture audio into text; read board pictures; read slides and teacher notes; identify important concepts and formulas; combine all information; find additional information when needed; generate organized lecture notes, examples, and practice questions.
Transition Model	When the agent processes each type of lecture material, the state changes. For example: raw lecture → audio converted to text → board information extracted → slides and notes processed → information combined → important concepts identified → complete lecture material generated.
Path Cost	The cost can be measured by the processing time, number of unnecessary or repeated steps, or amount of extra information processed. The objective is to generate useful lecture material with fewer unnecessary steps.
Goal Test	The goal is reached when the lecture has been completely processed and the agent produces organized lecture-wise material containing important concepts, explanations, formulas, examples, and practice questions.
State Space	All possible states of the lecture material, such as raw lecture, processed audio, extracted board information, processed slides, combined information, identified concepts, and completed lecture material.
Search Space	All possible sequences of processing the audio, board pictures, slides, and notes, combining the information, identifying important content, and generating the final lecture material.
Agents	Multiple agents. The AI Lecture Companion works with the student and teacher. The teacher provides the lecture material, while the student uses the generated lecture material for learning.

2. Environmental Properties

Property	Classification	Why
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Observability	Partially Observable	The agent can only access the information provided through audio, pictures, slides, and notes. It cannot directly know everything the teacher explains or what the student understands.
Determinism	Stochastic / Non-deterministic	The same lecture material may sometimes produce different results because audio quality, images, or information can be unclear.
Episodes	Sequential	Earlier processing affects later steps. For example, extracted lecture text is used when creating the final notes.
Change	Dynamic	New information can be added while the agent is working, such as additional notes, board pictures, or lecture content.
Values	Discrete and Continuous	Some information is discrete, such as topics, formulas, and questions, while audio and image information can have continuous values.
Agents	Multi-Agent	The environment includes the AI agent, teacher, and student. They have different roles in the lecture process.

3. Choosing the State Type

Option	Suitability	Explanation
Multi-state	Most suitable	The agent passes through many states while solving the problem. It starts with raw lecture material, processes the audio and images, combines the information, identifies important concepts, and finally generates complete lecture material.

Final Answer

The **AI Lecture Companion Agent** is best classified as a **Multi-state agent** because it passes through several states while processing a lecture. It starts with unprocessed lecture material and moves through different stages such as audio processing, board-image processing, information combination, concept identification, and final material generation. It is also **partially observable, stochastic, sequential, dynamic, and multi-agent**. These properties describe how the agent works within the lecture environment.